

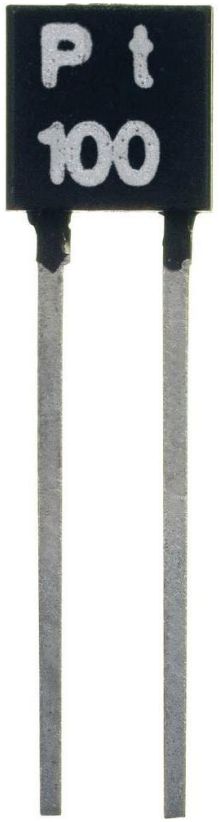
POLITECNICO
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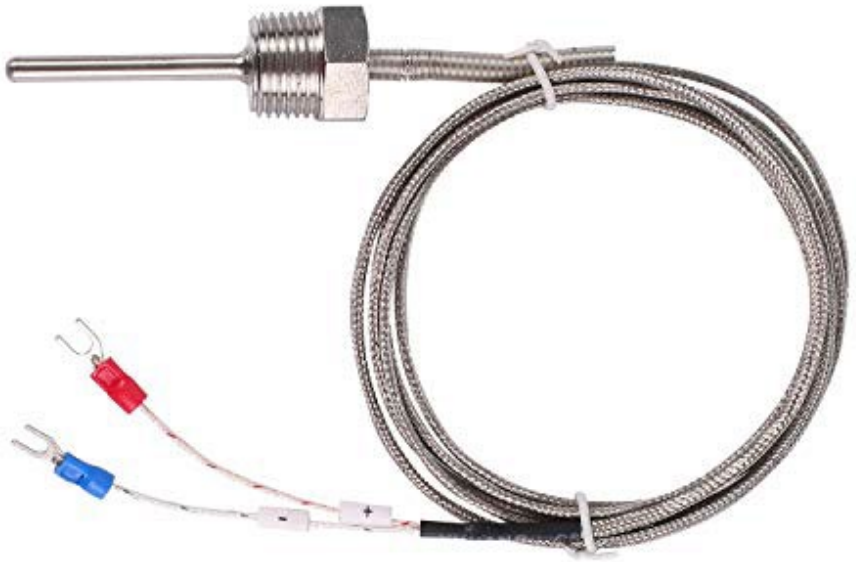
Temperature sensors
Part 1

SENSOR SYSTEMS

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- Resistance Temperature Detectors
- Thermistors
- Thermocouple
- Diode and bandgap temperature sensors
- Infrared thermometer





WeBeep “Supplementary materials”: 06 – Resistance Temperature Detectors

An RTD is a device which contains a metallic electrical resistance (referred to as a “sensing element” or “bulb”) which changes resistance value depending on its temperature.

This change of resistance with temperature can be measured and used to determine the temperature of a material.

Resistance: 100 Ω, 200 Ω, 500 Ω, 1000 Ω (Tolerances ±0.05% - ±0.1%)
Materials: Platinum, Nickel, Copper, Iron, Silver, Gold
Most common RTD are **100 Ω Platinum** sensors (Pt100).

Temperature range: -200°C – 800°C

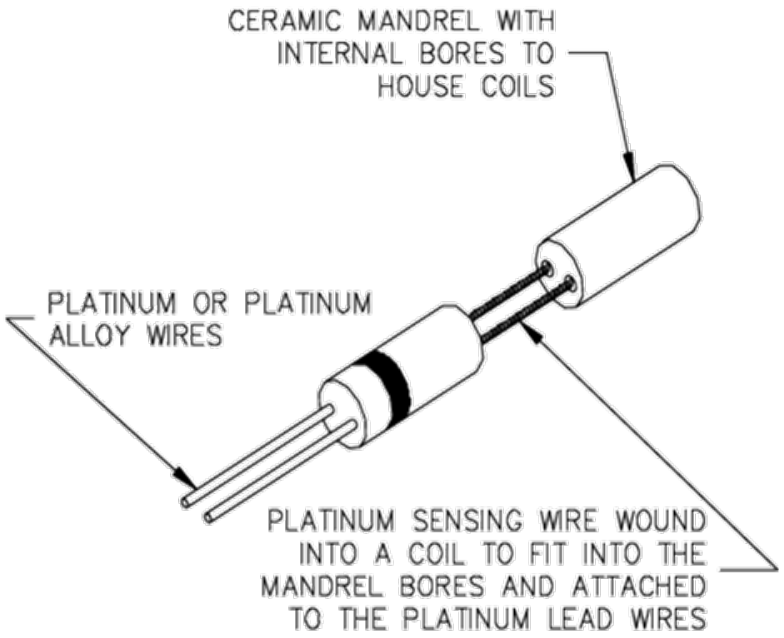
RTD sensing elements come in two basic structures

- Wire wound
  - Coiled design
  - Outer wound design

- Film pattern

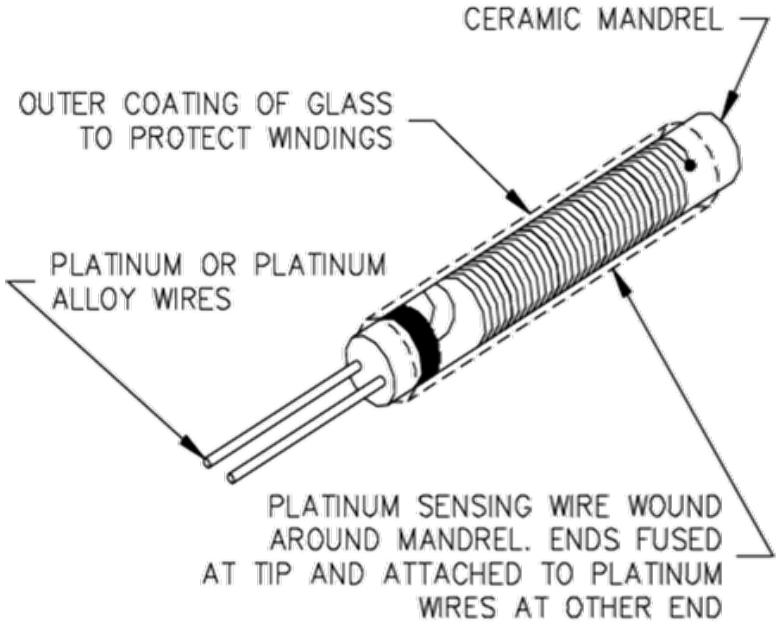
Wire wound elements contain a very small diameter metallic wire

(typically, 0.5 – 1.5 mil diameter, 1mil = 0.0254 mm).



WIRE WOUND ELEMENT – COILED DESIGN

wound into a coil and
packaged inside a ceramic mandrel



WIRE WOUND ELEMENT – OUTER WOUND DESIGN

wound around the outside of a ceramic housing
and coated with an insulating material

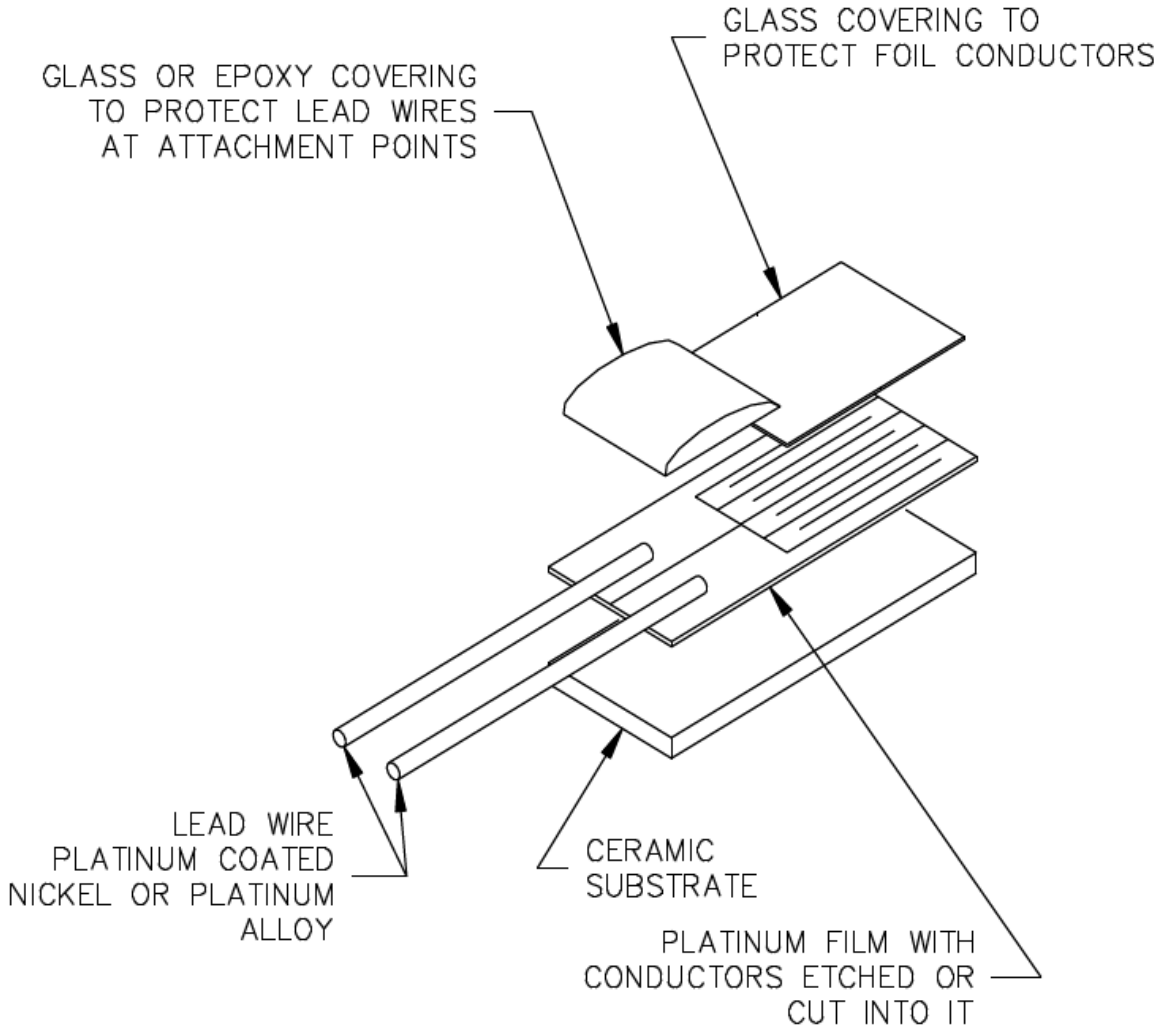
Film pattern

Film type sensing elements are made from a metal coated substrate which has a resistance pattern cut into it.

This pattern acts as a long, flat, skinny conductor, which provides the electrical resistance.

Lead wires are bonded to the metal coated substrate and are held in place using a bead of epoxy or glass.

Fabricated through photolithography



FILM STYLE ELEMENT

Sensitivity = quantity characteristic of a specific RTD

$$S = \frac{\Delta R}{\Delta T} \left(\Omega/_{\circ C}\right)$$

Temperature coefficient = quantity characteristic of a specific material:

$$\alpha = \frac{R_{100} - R_0}{100°C \cdot R_0} = \frac{S}{R_0} (\Omega/\Omega/°C)$$

RTDs typically have Positive Temperature Coefficient (**PTC**)

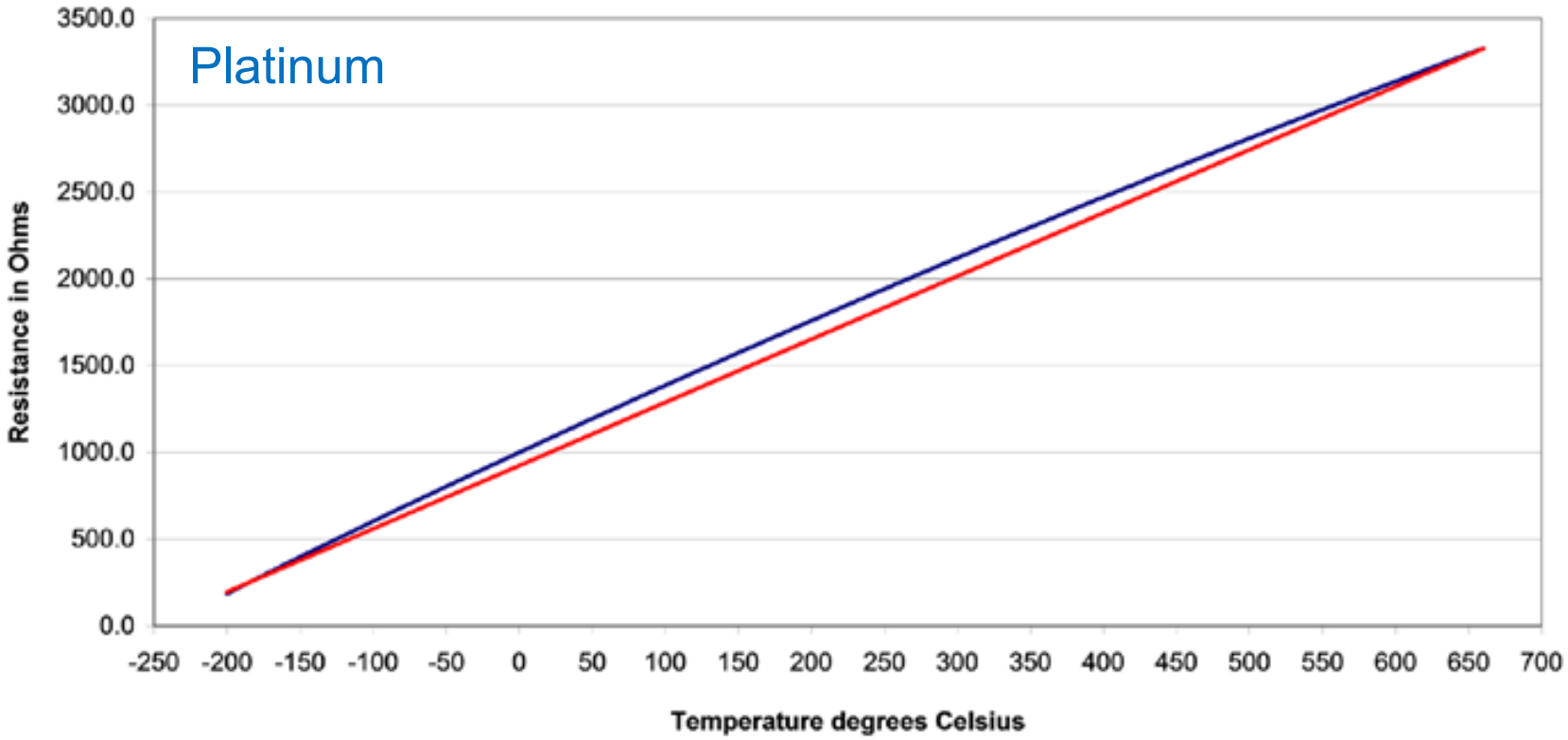
Characteristic equation:

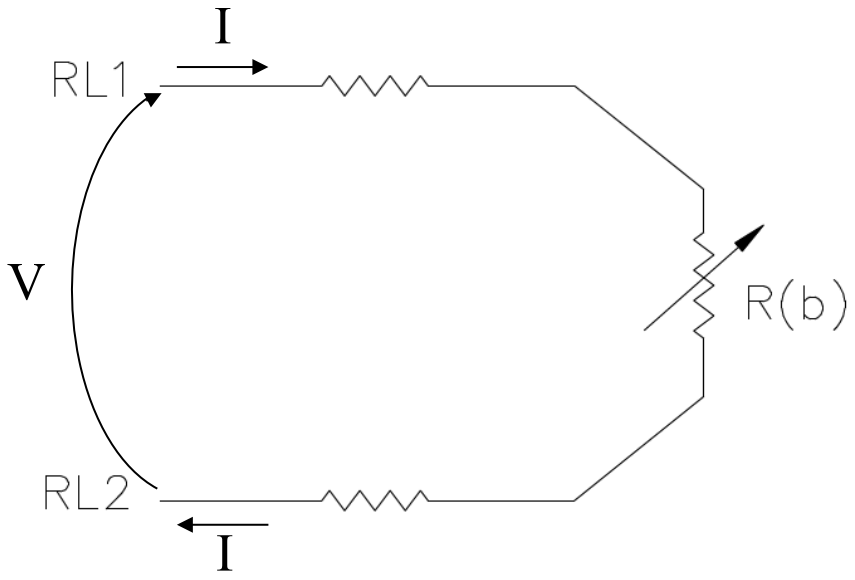
$$R_{RTD} = R_0(1 + \alpha\Delta T) = R_0 + \Delta R$$

$$\Delta R = \alpha R_0 \Delta T$$

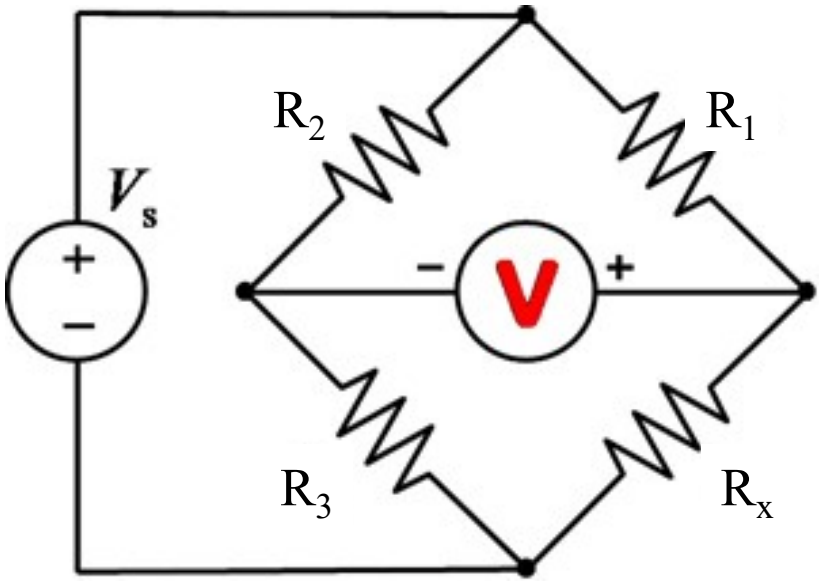
RTD temperature coefficient

Material	α
Platinum	0.0039 Ω/Ω/°C
Nickel	0.006 Ω/Ω/°C
Copper	0.0039 Ω/Ω/°C
Iron	0.005 Ω/Ω/°C
Silver	0.0038 Ω/Ω/°C
Gold	0.0034 Ω/Ω/°C





- 2 lead wires set-up
- 3 lead wires set-up
- 4 lead wires set-up



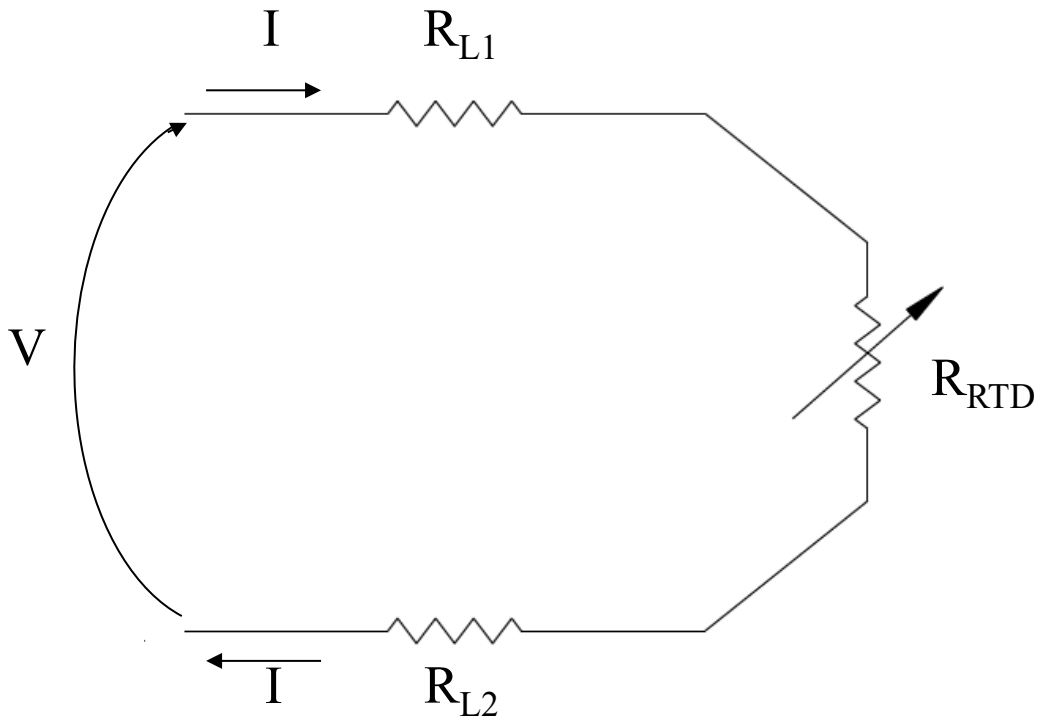
- Wheatstone bridge
- 3-wire bridge

2 lead wire set-up

It is the **least accurate**
since there is no way of eliminating the lead wire resistance from the sensor measurement.

2-wire RTD's are mostly used

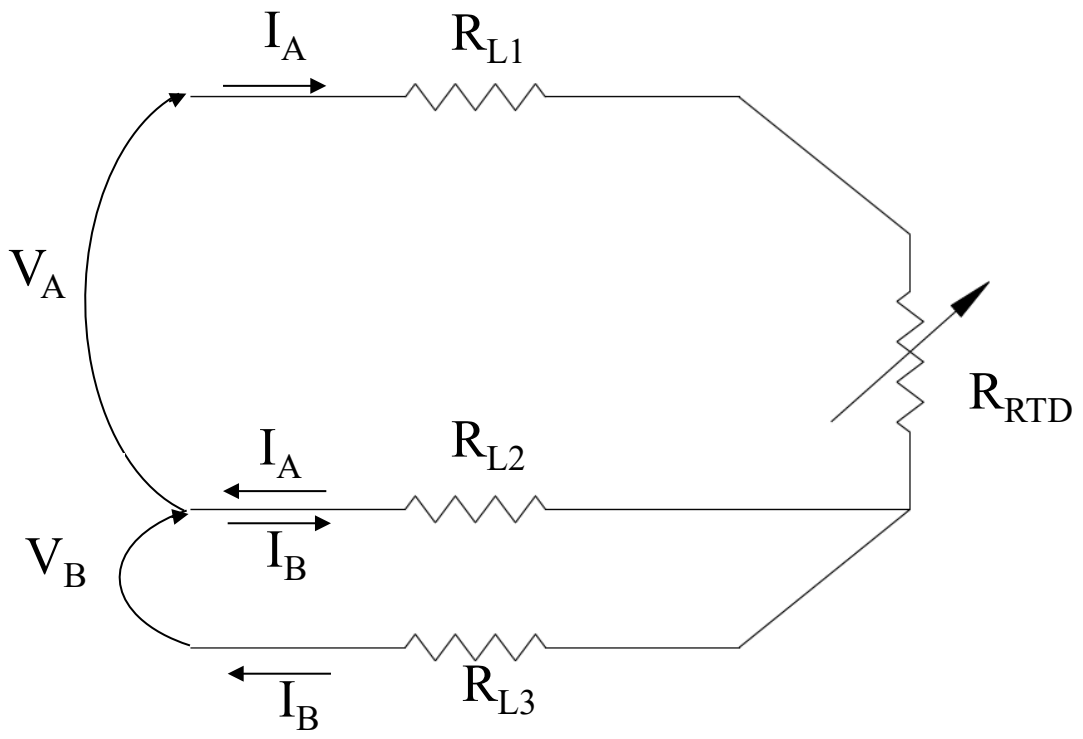
- with short lead wires
- where close accuracy is not required.



$$R_{meas} = R_{L1} + R_{L2} + R_{RTD}$$

It is the most used in **industrial applications**
where the third wire provides a method for removing the average lead wire resistance from the sensor measurement.

When long distances exist between the sensor and measurement/control instrument, significant savings can be made in using a three-wire cable instead of a four-wire cable.



$R_A = R_{L1} + R_{L2} + R_{RTD}$

$R_B = R_{L2} + R_{L3}$

$R_{meas} = R_A - R_B = R_{RTD}$

Assuming $R_{L1}=R_{L2}=R_{L3}$

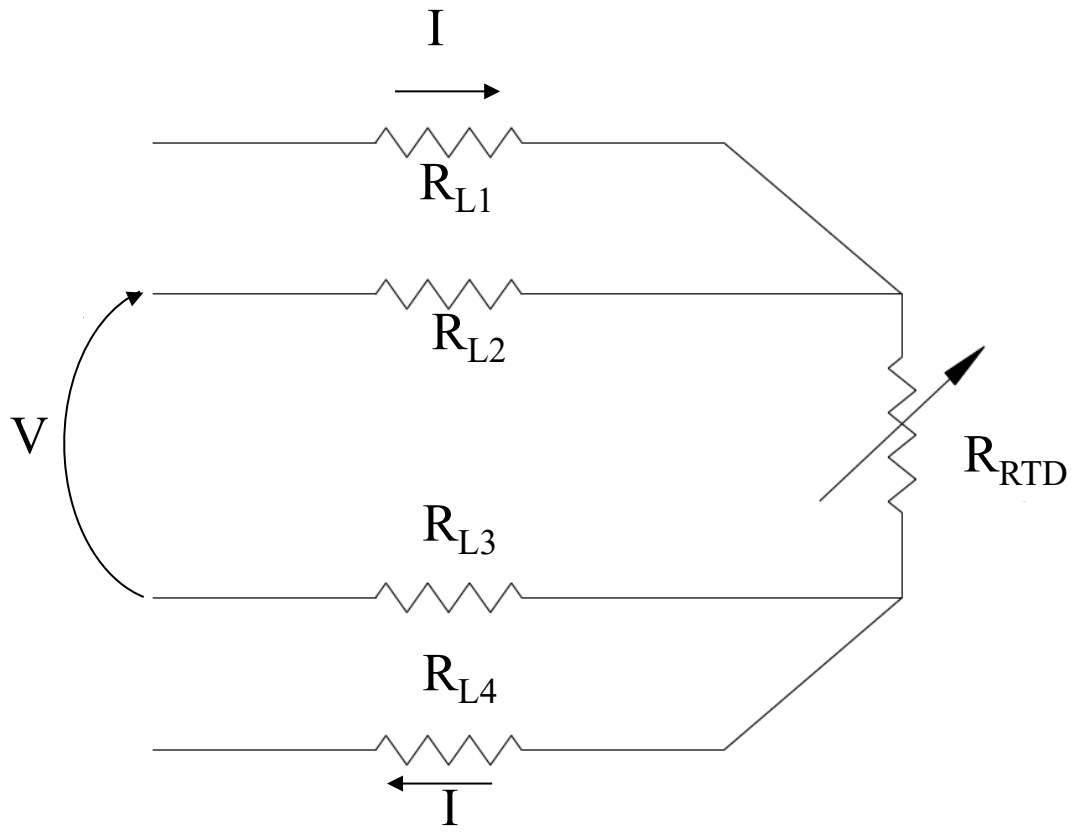
4 lead wire set-up

It is used primarily in **laboratory** where close accuracy is required.

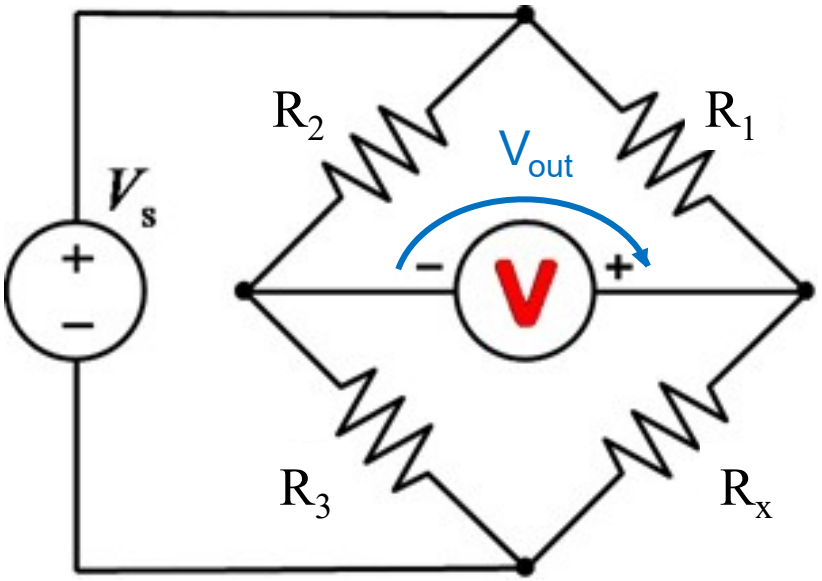
In a 4 wire RTD the actual resistance of the lead wires can be removed from the sensor measurement.

This method compensate for any differences in lead wire resistances.

The input impedance of the voltage measurement circuitry must be high enough to prevent any significant current flow in the voltage leads.



$$R_{meas} = R_{RTD}$$



R_{x0} nominal value of R_x → $R_x = R_{xo} + \Delta R$

In order to have V_{out}=0 when $R_x = R_{x0}$:

$$\frac{R_2}{R_3} = \frac{R_1}{R_{x0}}$$

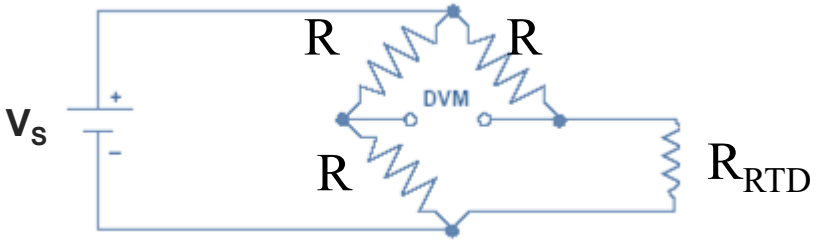
Considering $R_1 = R_2$ and $R_3 = R_{x0}$:

$$V_{out} = V_s \cdot \left[\frac{R_{xo} + \Delta R}{R_1 + R_{x0} + \Delta R} - \frac{R_{xo}}{R_1 + R_{x0}}\right] = V_s \cdot \frac{R_1}{(R_1 + R_{x0})^2} \cdot \frac{\Delta R}{1 + \frac{\Delta R}{R_1 + R_{x0}}}$$

$$V_{out} \approx V_s \cdot \frac{R_1}{(R_1 + R_{x0})^2} \cdot \Delta R = V_s \cdot \frac{R_1}{(R_1 + R_{x0})^2} \cdot \alpha R_{x0} \Delta T$$

Choosing $R_1 = R_2 = R_3 = R_{x0}$ provides the maximum $\frac{V_{out}}{\Delta T}$ → $\mathrm{V_{out}} = \frac{1}{4} \cdot \frac{\Delta R}{R_{x0}} \cdot V_s$

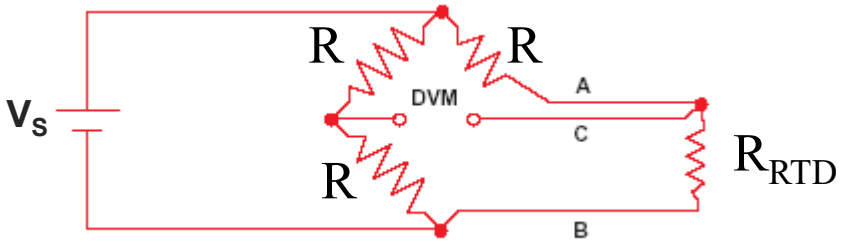
2-wire



Ideally ($R_L = 0$):

$$V_{out} = V_S\left(\frac{R+\Delta R}{2R+\Delta R} - \frac{1}{2}\right)$$

3-wire



Considering $R_{L1}=R_{L2}=R_L$

In 2-wire bridge:

$$V_{out-2w} = V_S\left(\frac{R+\Delta R+2R_L}{2R+\Delta R+2R_L} - \frac{1}{2}\right)$$

$$\varepsilon = V_{out-2w} - V_{out} =$$

$$= V_S \frac{2R_L \cdot R}{(2R+\Delta R)^2\left(1+\frac{2R_L}{2R+\Delta R}\right)} \approx V_S \frac{R_L}{2(R+R_L)}$$

In 3-wire bridge:

$$V_{out-3w} = V_S\left(\frac{R+\Delta R+R_L}{2R+\Delta R+2R_L} - \frac{1}{2}\right)$$

$$\varepsilon = V_{out-3w} - V_{out} =$$

$$= V_S \frac{-R_L \cdot \Delta R}{(2R+\Delta R)^2\left(1+\frac{2R_L}{2R+\Delta R}\right)} \approx -V_S \frac{R_L}{2(R+R_L)} \cdot \frac{\Delta R}{2R}$$

Since RTDs are resistors, they will produce heat when a current is passed through them.

The normal current limit for industrial RTDs is **1 mA**.

- Thin film RTDs are more susceptible to self-heating so 1 mA should not be exceeded.
- Wire wound RTDs can dissipate more heat so they can withstand more than 1 mA.

The larger the sheath or the more insulation there is the higher the error caused by self heating.

In some applications self-heating is exploited as an advantage.

RTDs pros and cons

Pros

- High linearity
- High accuracy
- Great repeatability
- High sensitivity
- Moderate price

Cons

- Narrow measuring range particularly at the high end
- Require an external power source
- Slow response time
- Self-heating

Typical applications:

- Air conditioning and refrigeration servicing
- Food Processing
- Stoves and grills
- Plastics processing
- Microelectronics
- Air, gas and liquid temperature measurement





WeBeep “Supplementary materials”: 07 – Thermistors

Thermistor = thermally sensitive resistor in which its primary function is to exhibit a change in electrical resistance with a change in body temperature.

Two basic types of thermistors:

- NTC thermistors = ceramic semiconductors (transition metal oxides), which decrease in resistance as the temperature increases.
- PTC thermistors = resistors generally made of polycrystalline ceramic materials, which increases in resistance as the temperature increases.

Typically, NTC or PTC thermistors do not increase or decrease in resistance linearly with temperature

Materials: metal oxides
(i.e., oxides of manganese, nickel, cobalt, iron, copper and aluminum)

Shape of the finished product: disc/chip, leaded/surface mounting device
(dictated by the specific applications)

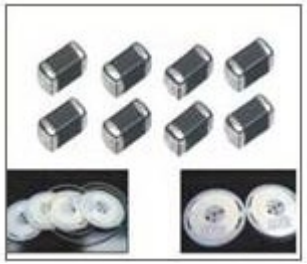
- Larger disc style NTC functions in the “self-heating” mode;
  that is, the change in resistance is a result of the wattage
  (heat developed by the passage of a relatively large current through the device).
- Smaller chip style NTC changes body temperature/resistance by absorbing the surrounding or ambient temperature.



disc

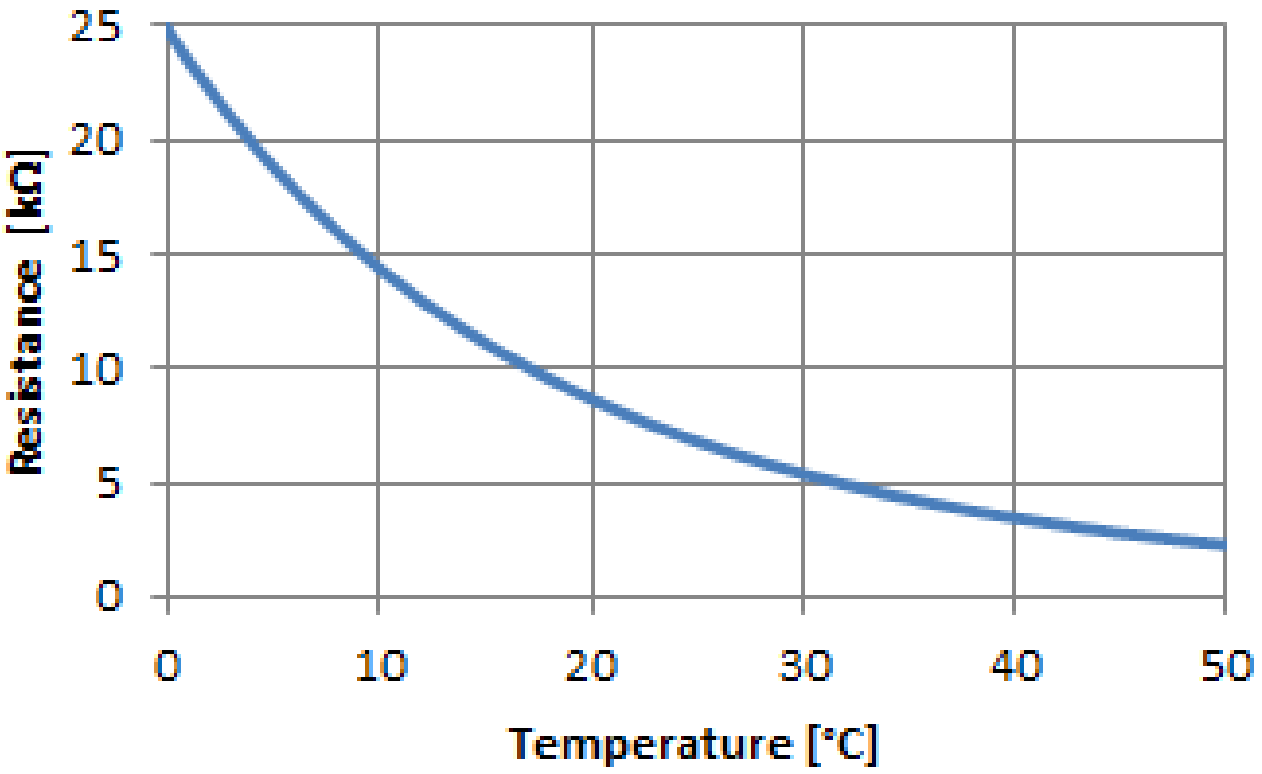


bead/chip



smd

NTC characteristic



$$R(T) = R_{25} \cdot e^{\beta\left(\frac{1}{T}-\frac{1}{T_{25}}\right)}$$

R_{25} resistance at 25°

β thermistor constant (depends on material)

T temperature expressed in Kelvin

T_{25} = 298.15 K

Temperature Measurement

Low-cost temperature measurement applications.

Temperature Compensation

Precision circuits, which require temperature compensation
(i.e., Oscillators, LCD displays, battery under charge and some amplifiers).

Fluid Level Applications

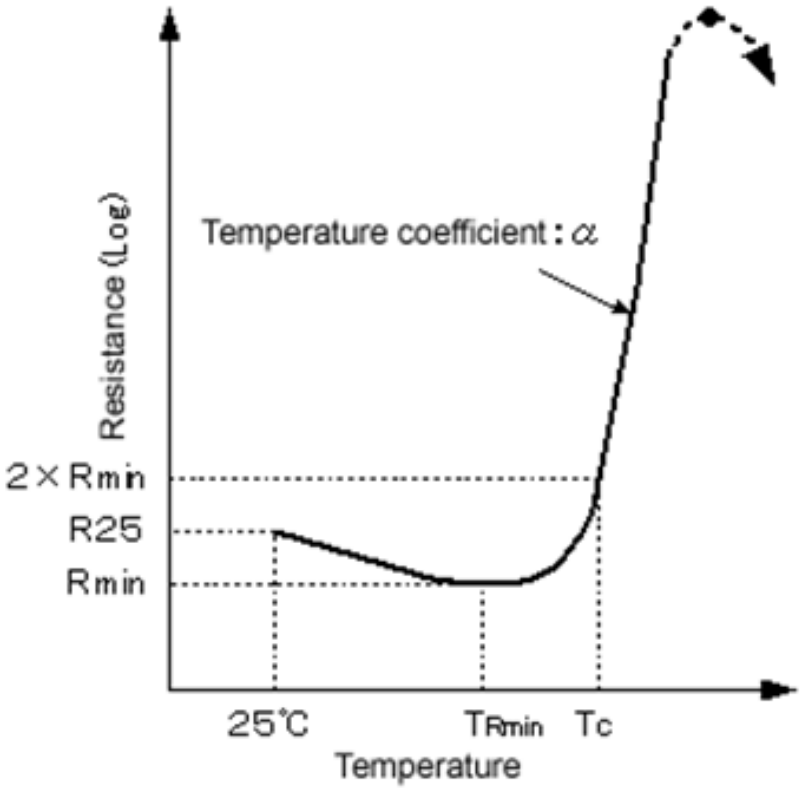
To sense the presence or absence of a liquid by using the difference in dissipation constants between a liquid and a gas.

Inrush Current Limiter

Disc NTC subjected to a change in power will experience a time lag before reaching a lower resistance.
This time lag can be utilized to limit the inrush surge current (the larger the part, the greater the lag).

Material: polycrystalline ceramic
(composed of oxalate or carbonate with added dopant materials)

The PTC thermistor exhibits only a slight change of resistance with temperature until the “switching point” is reached at which point an increase of several orders of magnitude in resistance occurs.



Tc = Curie temperature
the temperature at which resistance becomes twice the minimum resistance (Rmin)

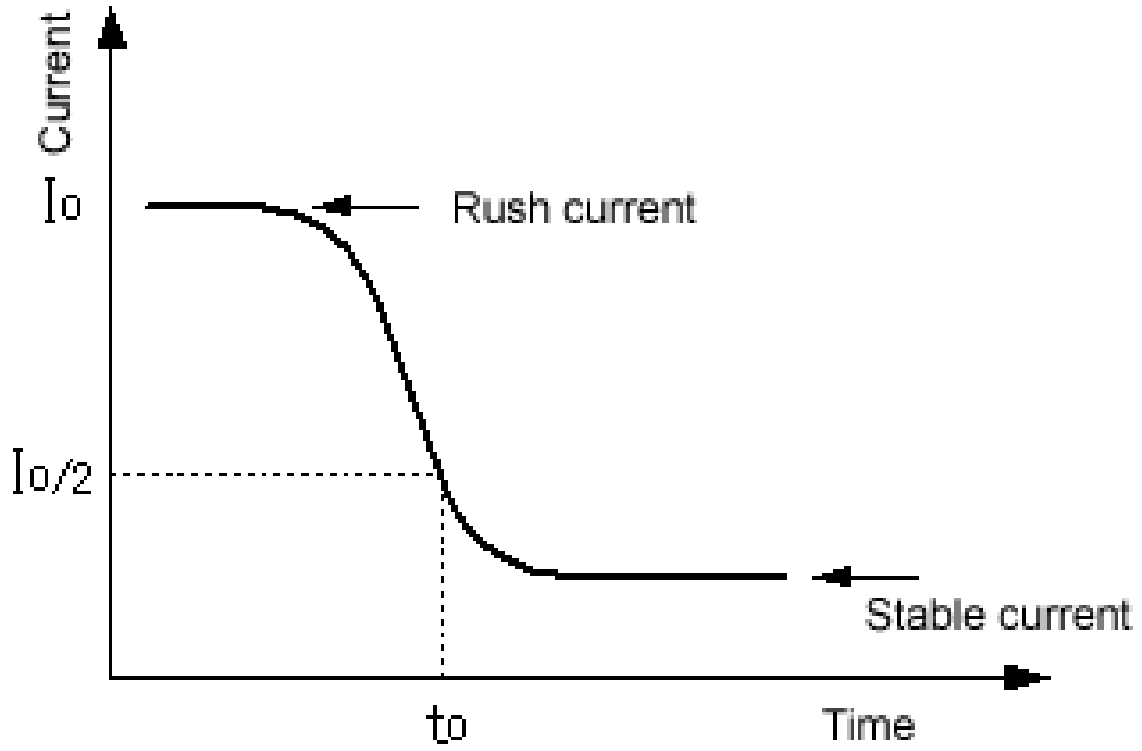
→Correspond to the switch point

Overcurrent Protection (Polyfuse)

When a fault condition occurs, PTC will heat up causing it to switch from a low to a very high resistance.

Battery Management

As a rechargeable battery becomes fully charged, its temperature increases and the RTD increases rapidly reducing the charge to a very low level.



Thermistors pros and cons

Pros

- High sensitivity (higher than RTD)
- Very moderate price
- Robust signal

Cons

- Very narrow measuring range (-100°C – 500°C)
- Low stability and linearity
- Medium accuracy
- Medium response time
- Self-heating

- Resistance Temperature Detectors
- Thermistors
- Thermocouple
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